

Persephone's Feet: A Foot-Based Approach to Play in Virtual Reality

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Figure 1: *Persephone's Feet*: a virtual reality game based on gardening entirely with foot-based gestures.

ABSTRACT

Persephone's Feet is a virtual reality game that questions: What if you could only garden with your feet? Exploring a series of three foot-based gestures (*Tap*, *Hover & Shake*, and *Kick*), this work presents a proof-of-concept for a novel virtual reality game for gardening. The game is designed as a relaxation and light exercise activity to take during breaks for the demographic of users who spend prolonged periods of time sedentary at a desk, which is becoming increasingly common for both work and education in modern society. A preliminary evaluation suggests that foot-based gestures can engender fun and motivate light exercise.

CCS CONCEPTS

• **Human-centered computing** → **User studies**; **Virtual reality**; **Gestural input**.

KEYWORDS

Virtual Reality, Games, Foot-based Gestures

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1 INTRODUCTION

Although room-scale Virtual Reality (VR) games typically rely on hand-controllers as opposed to foot-controllers, non-VR games that apply foot-controllers or other means of foot-based gestures have existed throughout video game history. The *Dance Dance Revolution*¹ arcade machine has floor pressure pads that detect dancing feet, and the *Wii Balance Board* [3] tracks a player's sense of balance for fitness games. *Hopscotch* is an age-old children's game based on hopping through a pattern of polygons and retrieving a tossed pebble. With VR however, sparse games depend entirely on foot-based play. Thus, this work proposes *Persephone's Feet*, a VR game based entirely on gardening with foot-based gestures. We thus investigate the following research questions:

- **RQ1:** How can different foot-based gestures for gameplay enhance user experience (respective of factors such as enjoyment and perceived workload)?
- **RQ2:** How can we develop a framework of foot-based gestures suitable for a VR gardening game?

2 RELATED WORK

Foot-based gestures refer to a predefined movement that triggers a specific interaction [4]. Research investigating foot-based gestures as an input modality evaluates how different mappings of function to foot-based gestures may be used to navigate, manipulate objects, and control system parameters [18]. They can be categorised into four groupings [4, 20]: *semaphoric*, simple signals and basic foot

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gestures, e.g., tapping; *deictic*, meant to point or touch a specific target, e.g., tapping on an object; *manipulative*, which changes an object's properties, e.g., changing an object's location by flicking it; and *implicit*, referring to gestures that do not directly target an object for interaction.

Foot-based gestures can be found in different VR conditions: seated, standing, and walking [4, 21]. The input modalities include the location and position of each individual foot, the inter-foot distance, foot-pressure, and foot direction [21].

Kicking, for instance, was investigated for mobile interactions for contexts where the hands are occupied but the feet are free to move [7] as well as for large touch interfaces where part of the screen near the ground is inconvenient to reach with hands [9]. Saunders and Vogel examines a combination of taps, jumps and kicks for foot-based interaction at a standing display to encourage physical activity for stationary desk-users [15].

Tapping was also investigated in mobile contexts [2] and head-mounted display (HMD) contexts [12], exploring complex selection and navigation functions using different permutations of gestures from each foot. Ankle rotation is another technique, and examines rotating and pivoting the feet for remote interaction [16].

From a more subtle perspective, some foot-based gestures use shifts of balance instead of physically lifting the foot off the ground. Sockswitch [6] investigates pressure-based plantar gestures, such as pressing the sole of one foot on the ground as an input modality.

Furthermore, foot-based gestures have the advantage of being compatible with other modalities, such as hands or gaze-based interaction [13] and head movement [17].

3 INTERFACE INNOVATION

The application of foot-based gestures for VR HMDs is a recent field of research. Muller et. al.'s *Mind the Tap* in 2019 is the first systematic investigation on foot-taps as an interaction method for HMD devices [12]. Additionally, this study – though applicable for general HMD contexts – was evaluated in augmented reality (AR) using the Hololens, as opposed to both VR and AR devices. ShoeSoleSense (2013) proposes a novel wearable foot interface for virtual and real environments, and proposes user studies as future work [11]. Fukahori et al.'s Sockswitch (2015), a wearable sock with pressure-sensors, takes this work a step further by investigating subtle plantar gestures for HMD devices [6]. To this new and emerging body of research, we contribute a preliminary evaluation of the foot-gestures *Kick* and *Hover & Shake* in a VR environment for the purposes of play, with *Tap* as a base condition. Although *Kick* has previously been examined in mobile [2, 7] and display screens [9, 15], we add to the body of existing research by evaluating kicking in VR environments using a VR HMD (Oculus 2). *Hover & Kick* is a novel narrative-led gesture derived from our pilot study, inspired by participants hovering and shaking their foot to e.g., sprinkle water on a plant. Importantly, compared to the aforementioned studies we are specifically interested in the more qualitative, phenomenological experience of executing foot-based gestures. In *Persephone's Feet*, we investigate how foot-based gestures benefit user experience in terms of enjoyment and physical demand.

4 CONCEPT DESIGN

Persephone's Feet is a gardening game based on foot-based gestures in VR. The game targets individuals who spend significant time on desk-work in indoor environments. Being in or seeing nature is naturally relaxing [10] and gardening itself can be physical engaging. However, here is the twist: interactions in *Persephone's Feet* are entirely based on feet gestures as opposed to the more typical hand-based interactions. The idea derives from the Greek myth of the goddess of spring, Persephone, whose footprints sprouted buds, flowers, and other beautiful things: '*as the Spring herself first flutters here and there, while close upon the delicate printing of her feet are found the new-born flowers and buds and fairest things.*' [5].

To test the boundaries of how far a foot-only based game can go, we focus on feet interaction alone. For our narrative setting, we selected an activity that that could occur within a stationary platform, like a small garden, as opposed to large football fields which may require additional techniques (e.g., teleportation) to relocate the player in order to enable play. Moreover, we wanted to explore a game scenario unique to the fictional reality of VR yet still relatable to the user, namely exerting the powers of the goddess Persephone for a mundane activity such as gardening.

To investigate what foot gestures would be intuitive, appropriate, and fun to execute as game mechanics, we conducted a pilot study with a paper prototype to identify the most commonly used foot gestures when participants faced the task: 'nurture your garden only using your feet'.

5 PAPER PROTOTYPE

A pilot study was conducted with 4 participants (1 female, 3 male) using paper prototypes via a Wizard of Oz approach [22]. The wizard, namely the researcher, mimicked the basic game mechanics of *Persephone's Feet* while the participants were free to move their feet in any way they liked to interact with prototype.

Figure 2 provides context for the paper prototype: The researcher first guided the participant to a demarcated circle on the ground, and asked the participant to stay within the circle. Although participants were allowed to extend their limbs beyond the boundary of the circle, they were not allowed to put their foot down on the ground beyond the boundary. A paper watering can was attached to the participant's preferred foot, and a paper fertilising bag was attached to the other. The researcher, simulating wind in the game, came

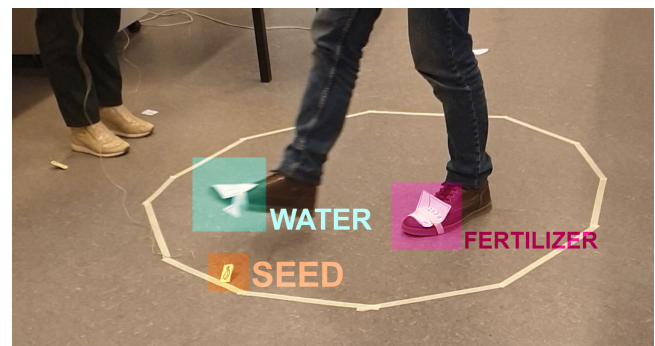


Figure 2: Paper Prototype

around and dragged seeds within and out of the circle using a piece of string. Participants 'caught' seeds with their feet, and watered or fertilised the seed by moving or shaking the appropriate foot. Once the researcher deemed the plant as sufficiently nurtured, the researcher tossed leaves on top of the seed as a representation of incremental growth. Occasionally, the researcher removed plants that had been neglected by the user as a sign of plant death from lack of care. The game ended once the participant nurtured a plant until it grew enough to bud a paper flower.

The three most commonly used foot gestures observed during the study were: (1) *Tap*: to plant a seed in the ground and prevent it from fluttering away; (2) *Hover & Shake*: to water or fertilise a plant; (3) *Kick*: swinging the foot forwards to splash water or fertiliser towards plants out of the participant's immediate reach. Other less commonly used foot gestures include hopping, jumping, or swinging feet laterally to reach plants outside the circle.

6 GAME DESIGN

Players of *Persephone's Feet* are placed in a virtual garden to nurture only with their feet. As seen in Figure 1, the virtual garden consists of three tiers: (1) the circular 1.5 m diameter platform within which the players have the agency to walk; (2) a tier below the platform which is not walkable, but players can extend their limbs beyond platform boundaries to access this space with limited reach; (3) a virtual beyond completely inaccessible to the player, but plants within this space will grow irrespective of the player's care to provide an indication of passing time whilst in the VR environment.

In terms of gameplay: Seeds are either found on the ground or flutter softly from the sky at random moments in time. Upon interaction, these seeds are 'planted' and transition into their growth stage. In the growth stage, the plants will indicate they need a specified component to grow further: water, flower fertiliser, and grass fertiliser. These three components (water, flower fertiliser, and grass fertiliser), exist as virtual tools on the ground that the player must interact with to 'attach' the tool onto the selected foot, e.g., interacting with the watering can with the left foot will turn the foot blue and enable watering interactions. A player can have a maximum of two different tools on feet to nurture the garden, thus must occasionally switch the attached tool as there are a total of three different plant needs. Throughout the growth stage, plants will continue to randomly signal different needs and grow incrementally when the need is satisfied; however, if the plant is neglected by the player for more than 30 s the plant will fade out and 'die'. Once the plant reaches full maturity, it will no longer grow and require further care. The specific foot gestures used to interact with the garden are described in the following subsection.

6.1 Foot gestures

The three most commonly used foot gestures from the pilot study were developed for the VR implementation of the game. All three foot gestures accomplished the same function: to nurture virtual plants in a VR garden. However, each gesture embodied a distinct area of influence for interacting with the plants. As seen in Figure 3, from smallest to largest area of influence:

- **Tap**: Defined as a collision between the foot and an artefact on the ground plane. The area of influence for triggering an

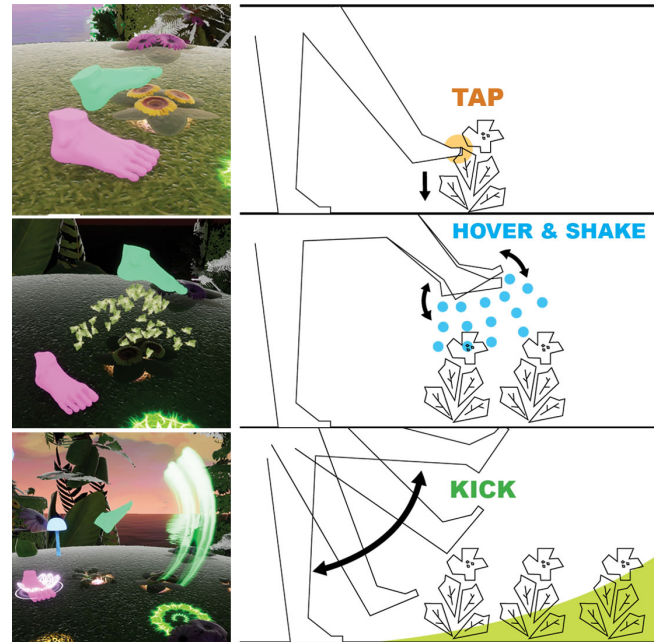


Figure 3: Three ways of gardening with feet: (1) *Tap*, (2) *Hover & Shake*, and (3) *Kick*.

interaction is constrained to the contact point between the foot and virtual artefact.

- **Hover & Shake**: Defined as short bursts of movement when the foot is suspended above ground. The area of influence is slightly larger, enabling interactions with multiple objects mostly underneath the foot.
- **Kick**: Defined as a swing of the leg from ground plant into the air (and vice-versa). It has the largest area of influence of the three foot gestures, enabling interactions with the chain of artefacts in the direction of the kick within 2.5 m distance.

When an interaction was triggered between a foot gesture and a virtual plant, both visual and haptic feedback were implemented to indicate the gesture had occurred. The virtual plant(s) interacted with glowed, while the controller used for the interaction vibrated.

6.2 Technical Details

The method for recognising the foot gesture carried out by the user was based on different thresholds for (1) the *height* of the foot above ground plane and (2) the *instantaneous speed* of the foot, calculated as the magnitude of the foot's velocity. To elaborate, if the height of the foot was lower than a threshold of 14 cm above ground plane, the programming identified the user's foot state as 'rest'. If the foot maintained this state and overlapped with a plant in the scene, the foot state would switch to 'tap'. In contrast, if the foot height exceeded this threshold, the programming identified the user's foot state as a 'hover'. While the hover state is maintained, the foot state switches to 'shake' if the instantaneous speed exceeded the threshold of 100 cm s^{-1} . If the height of the foot exceeds an even higher threshold of 35 cm above ground plane and the instantaneous speed exceeded 20 m s^{-1} , the gesture would be classified as a 'kick'.



Figure 4: Chronological narrative of securely attaching VR hand controllers onto a human foot.

Regarding the physical setup: The game employs a VR head-mounted display (VR HMD), a pair of controllers, and a pair of makeshift foot-straps made of cloth strips and Velcro. As portrayed in Figure 4, a specific strapping method was designed to securely attach the VR controllers to a user's feet without impeding gait or restricting foot movement.

7 PROOF-OF-CONCEPT

Following the pilot study with the paper prototypes, another study was conducted with the virtual proof-of-concept of *Persephone's Feet*. The game was evaluated in respect to the three foot gestures implemented for gardening, namely (1) *Tap*, (2) *Hover & Shake*, and (3) *Kick*. Participants were asked to nurture a plant to full maturity using any or all of these foot gestures, after which they were free to end the VR simulation or continue to nurture other plants in the game. No time limit was given to complete this task.

7.1 Procedure

The researcher invited the participant to read a consent form and an information sheet describing the context of the study. After obtaining consent, the researcher requested the participant to sit and remove their shoes to attach the pair of controllers onto each respective foot. The participant was then asked to put on the VR HMD, placing the participant into the virtual environment (VE).

While the participants were inside the VE, the researcher tasked the participant with nurturing at least one virtual plant, informed the participant on the general game mechanics, and asked the participant to attempt all three foot gestures at least once. Afterwards, the researcher no longer intervened and the participant played the game until task completion. Participants were allowed to continue playing even after task completion if so desired.

After the experiment, the researcher asked the participant to complete a series of post-experiment questionnaires: Intrinsic Motivation Inventory (IMI) for subscales 'Interest/Enjoyment' and 'Competence', the NASA Task Load Index (NASA TLX), and the Slater-Usch-Steed (SUS) Presence questionnaire. Finally the participant was asked to participate in a 3 min semi-structured interview.

7.2 Participants and Apparatus

A total of 5 (2 female, 3 male) participants took part of the study, with a mean age of 30.2 years ($SD = 6.76$). Participants rated their experience with immersive technologies (e.g., VR, AR, mobile-based) on a 7-point scale from no to expert experience ($M = 4.00$, $SD = 2.12$). The participants wore an Oculus Quest 2 HMD and strapped a pair of motion controllers onto their feet. Figure 4 illustrates the method of securely attaching the motion controllers. The game was exported into Oculus Quest 2 as a standalone executable, and the experiment was carried out through wireless operation. The experiment took place in different locations, each with a minimum of $3\text{ m} \times 3\text{ m}$ clear floorspace.

8 RESULTS

We analysed our questionnaire data (IMI, NASA TLX, and SUS) and conducted a thematic analysis [1] on the interviews. Participants' sense of presence, namely the 'sense of being' in a VE, was measured using the Slater-Usch-Steed (SUS) Presence questionnaire [19]. Based on a 7-point Likert scale, presence scores were calculated by counting the number of responses marked greater than 5. Participants generally felt that the VR experience elicited a normal sense of being in a place. Results showed a presence count score of 3.40 ($SD=3.4$) with an average presence score of 5.37 ($SD=0.73$).

The subscales 'Interest/Enjoyment' and 'Competence' of the Intrinsic Motivation Inventory (IMI) [14] were used to measure the entertainment value of the game as well as game balance. Participants generally enjoyed the game (interest: $M=5.91$, $SD=0.501$). the perceived competence converged to average (competence: $M=4.00$, $SD=1.33$), suggesting that users felt the game was challenging but not overly so. A Shapiro Wilk test shows that neither subscales are normally distributed (interest: $p=0.0013$; competence: $p<0.001$), as illustrated in Figure 5.

The NASA Task Load Index (NASA TLX) is a multidimensional assessment tool to measure the perceived workload of a task [8], or in this case, gardening. The NASA TLX is scored out of a scale of 100, where 0 is no perceived workload and 100 represents very high workload. As illustrated in Figure 6, from lowest to highest, the workloads for the following six categories are: Frustration ($M=28.6$, $SD=19.17$), Mental Demand ($M=30.2$, $SD=18.16$), Effort ($M=40.4$, $SD=17.13$), Temporal Demand ($M=44.2$, $SD=20.39$), Performance ($M=57.8$, $SD=14.79$) and Physical Demand ($M=59.4$, $SD=14.8$).

8.1 Interviews

From the interviews, we identified 3 common descriptors for the foot gestures: Intuitive, Fun, and Physically Exerting. Regarding notation, '(n)' is used as a shorthand to refer to the number of participants who cited such a statement.

Intuitive. 3 participants mentioned *Tap* as the most intuitive form of foot-based gesture, while 2 highlighted *Hover & Shake*. However, the reasons for describing why the gesture felt intuitive differed. For example, "*Tapping was intuitive. It made sense. It was an efficient, normal technique to use.*" in contrast to "*Hovering & Shaking felt the most intuitive, because you are spreading seeds or water*". *Tap* seems to be intuitive due to the simplicity of the gesture, while *Hover & Shake* seems to be intuitive from a narrative perspective.

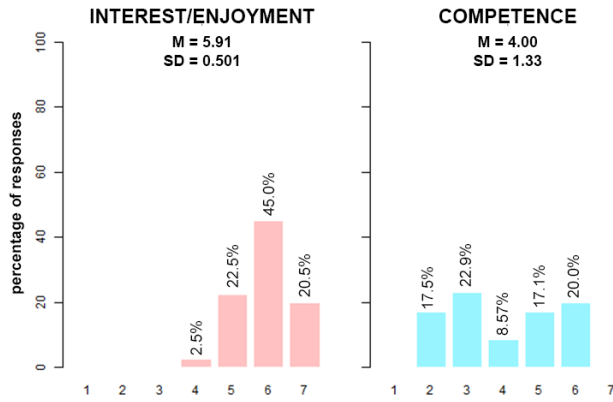


Figure 5: IMI (Interest & Competence)

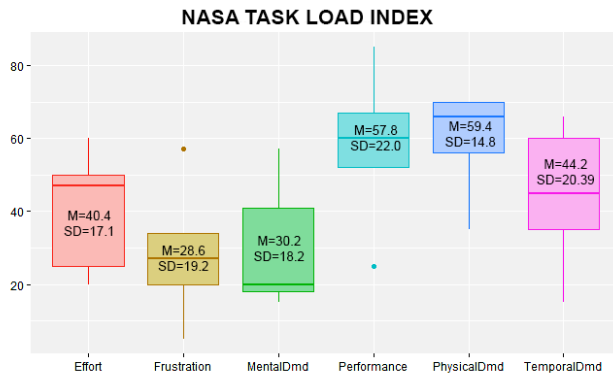


Figure 6: NASA Task Load Index, Unweighted

Fun. 3 participants highlighted *Kick* and 1 mentioned *Hover & Shake* as the most fun gesture. For instance, “*Kicking was fun. I liked to see water coming out of my feet.*”. No participant indicated that “*Tap*” as the most fun. However, *Kick* (1) and *Hover & Shake* (1) were both cited as somewhat ‘scary’, e.g., “*It was a bit scary to kick outside the platform because it felt like I would fall.*”. Nevertheless, the feeling of fear for both reasons were tied to the VE rather than the gesture itself, thus adjusting the appearance of the environment could potentially alleviate the fear of falling.

Physically Exerting. 1 participant cited *Kick* and another mentioned *Hover & Shake* as physically exerting. However, the participant who cited *Kick* as tiring also cited *Kick* as both their favourite and most frequently used technique: “*Kicking was the most fun...I used it most often because it was easier...I was even kicking for the nearby plants because it was faster.*”. In contrast, the participant who cited *Hover & Shake* indicated that the technique was tiring due to the difficulty in maintaining balance on one foot in order to hover and shake the other.

9 DISCUSSION

The aims of this paper are to investigate how different foot-based gestures can enhance gameplay and user experience (**RQ1**) and evaluate a framework of foot-based gestures in terms of their suitability for a VR gardening game (**RQ2**). A preliminary evaluation suggests that foot-based gestures can be used to develop a VR game that is both enjoyable and reasonably challenging, as well as physically exerting yet mentally easygoing.

In response to **RQ1**, findings from the interviews show that *Kick* was most frequently cited as the most fun method of interacting with the VE, followed by *Hover & Shake* and lastly *Tap*. The opposite trend is perceived for how intuitive a foot-gesture felt to execute, with *Tap* being the most natural and easy-to-learn gesture. This indicates that the sensation of ‘fun’ is not necessarily correlated with the intuitiveness of a foot-gesture.

In terms of the perceived workload of the task: Frustration followed by Mental Demand are the lowest perceived factor for workload in the game while Physical Demand and Performance were the highest contributing factors of the six categories identified in the NASA TLX (see Figure 6). Interviews substantiate these results. One user highlighted the immersive and tranquil quality of the game, “*I could see myself drawn into the surroundings and being absorbed into the ambience*”; and many affirmed that they “*felt like I moved around*” in the VR simulation. This falls in line with the motivations of the game - to create a relaxing activity that provides light exercise for individuals with desk-jobs.

General observations by the researcher during the user study suggest improvements for each foot-based gesture. *Kicking* could have improved visualisation to more clearly show its area of effect for interaction, e.g., “*It would be nice to have a visualisation of what I am kicking so that I can control my gesture more.*” *Hover & Shake* could be better implemented to show the same, e.g., “*Hover & Shake was the most fun but not very accurate*”. No suggestions were made for *Tap*, but due to its simplicity perhaps *Tap* can be used as a basis for comparison with other more complex techniques.

In response to **RQ2**, preliminary results indicate that participants generally regarded the gestures used in *Persephone's Feet* as suitable for gardening. The responses from the SUS questionnaire indicate that participants typically felt present in the game despite the fact that using feet is an atypical way of gardening. However, although the mean perceived competence met the goal of game balancing (4 in the 7-point scale), the Likert distribution was not normal (see Figure 5) signifying that some users found the game easier to play than others. Nevertheless, the game balancing is less related to the foot-gestures themselves and more related to other aspects of game mechanics. For instance, one user suggested implementing “*some goals in the game to keep you motivated, e.g., you need to raise 5 flowers, to keep you motivated*”.

The interviews also provided insights for the suitability of the game for its intended case as a relaxing and light exercise activity to take during breaks from desk work. The participants generally agreed: e.g., “*I imagine it as a gardening game where I take my time to explore*”, or “*I think it can be useful. I don't see why not, it can help if you're in the office many hours and you need exercise but not extreme exercise.*”. Interestingly, one participant suggested that the game can be adapted for a different use case as an exhilarating

exercise VR game: “I was thinking you could do the opposite (of light exercises well, like a Beat Saber of growing things).”

9.1 Limitations & Future Work

The game was only tested with a small sample size (5 participants), which provides a preliminary evaluation but perhaps not a holistic one. If the issue of a small sample size persists for future work, additionally implementing focus groups could reveal a more detailed examination of participants' qualitative experience.

Moreover, all the tested foot gestures were combined into a single experience, making it difficult to compare how effective each gesture was in affecting user experience. A future work comparing an improved implementation of the gestures above, perhaps with additional interactions informed by the pilot study such as jumping or hopping, could provide a deeper insight into the foot-based gesture component of the VR game. A more rigorous study could also include a comparison between two conditions, foot-based gestures and hand-based gestures for the implemented gardening effects.

Future studies could also further investigate Persephone's Feet as a light exercise activity or exergame by exploring quantitative exertion metrics such as heart rate, approximate calorie expenditure, etc. In such a study, we could pose hypotheses investigating either the performance of a specific foot gesture for exercise or the performance of foot-gestures for a range of activities. Additionally, to further explore the design space of foot-based interactions for play, we could investigate a physics-sensitive approach, such as using the magnitude of a foot's velocity to influence the extent to which an interaction affects the VE.

10 CONCLUSION

Persephone's Feet presents a novel gardening game based entirely on foot-based gestures instead of the standard hand-based controls for engaging in VR gameplay. The game is designed as a relaxation and light exercise activity to take during breaks. A preliminary evaluation suggests that foot-based gestures can engender fun and motivate light exercise, and may be used to develop VR gardening game that is reasonably challenging yet relatively mentally enjoyable. The three foot gestures developed in this study (*Tap*, *Hover & Shake*, *Kick*) has the potential to complement other modes of interaction for VR gameplay. For instance, with hand-controller based gameplay, the additional feet controllers enables the user to have multiple points of interaction with the VE. Moreover, if these three foot gestures can be used for VR gardening, the gestures also have the potential to be applied for other genres of play in VR games. *Persephone's Feet* thus presents just one of many ways to provoke fun and exercise using foot-gestures in VR.

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